



sTMS Interface Functional Specification

Singapore Polytechnic – SG Secure System

Version 1.0

Document Version

Version	Effective Date	Summary of Changes	Created/ Updated By
0.1	29/03/2022	Initial version of SG Secure System IFS	AvePoint
0.2	23/04/2022	Revised based on workshop feedback from SP. Update 2.1. System Flow diagram and description, sTMS will push file to SP SFTP server; Update 5. Interface Sample Files	AvePoint
0.3	16/05/2022	Revised based on email feedback from SP. Change the transfer file extension from ".csv" to ".txt". Remove the assumption, that SP CET student who has the CEPAS number will be transferred to SG Secure system. Document Acceptance Form Update Approved By to SP PIT lead	AvePoint
0.4	13/06/2022	Revised based on email feedback from SP. - Update description of 1.1 Purpose of This Document. - Update error handling of 2.2 Interface design summary. - Update 2.3.2 Business Rule: use the sTMS terminology instead of using the SP TMS Terminology; Revised term based on stage1 & stage2 fs design; Rename Course Series No. to SP Existing TMS Course Careers.	AvePoint
0.5	12/09/2022	Revised based on meeting feedback from SP. - Update the system flow diagram and description; - Remove CEPAS No. and revised the business rule 4&5; - Update the course category and course type table; - Update SP SG Secure System scenario and sample files.	AvePoint
1.0	16/09/2022	Official version for acceptance and sign-off.	AvePoint



Document Acceptance Form

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Table of Contents

- Document Version 1
- Document Acceptance Form 2
- 1. Introduction 2
 - 1.1 Purpose of This Document 2
 - 1.2 Intended Audience 2
- 2. Technical Overview 3
 - 2.1 System Flow 3
 - 2.2 Interface Design Summary 4
 - 2.3 Business Flow 4
 - 2.3.1 Flow to Generate SG Secure TXT File 4
 - 2.3.2 Business Rule 5
- 3. Interface Fields 7
- 4. Interface Sample Files 8

1. Introduction

1.1 Purpose of This Document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirement and detailed technical specification between the SaaS shared Training Management System (sTMS) and Singapore Polytechnic SG Secure System to transfer students access card details.

sTMS shall identify students' enrollment status based on the course type, generate corresponding TXT file for these students, and deposits the TXT file encrypted by PGP protocol in the SFTP server accordingly. Singapore Polytechnic can use the transferred TXT file for the turnstile access.

1.2 Intended Audience

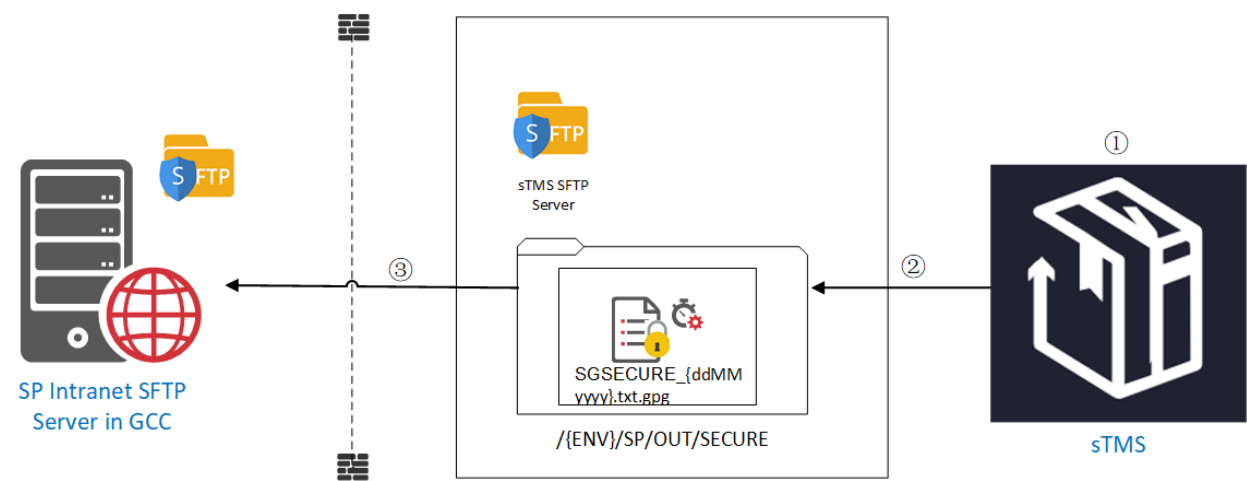
This technical document is used for system communication. Therefore, the intended audiences are the technical stakeholders of both systems.

2. Technical Overview

2.1 System Flow

There are 3 roles involved in the interface diagram.

- **SP Intranet SFTP Server:** Singapore Polytechnic Intranet SFTP Server in GCC.
- **sTMS SFTP Server:** A SFTP server hosted in Azure that stores encrypted TXT file used for data exchange.
- **sTMS:** shared Training Management System.



SN	Description
1	sTMS generates the SG Secure TXT file based on student's enrollment status on a pre-defined schedule.
2	sTMS configures Timer job to push SG secure info file into sTMS SFTP server after encryption with PGP protocol.
3	SP pulls the SG Secure TXT file from the sTMS SFTP server on demand.

2.2 Interface Design Summary

Output Interface File

Interface File Name	SGSECURE_{DDMMYYYY}.txt.gpg
Data Source System	shared Training Management System
Data Destination System	Singapore Polytechnic SG Secure System
Mode of Interface	TXT file, with “ ” as delimiter
Method	SFTP
File Path	/{ENV}/SP/OUT/SECURE ENV: UAT, PROD
Encryption Method	PGP
Data Syncing Mode	full load data sync
Frequency	Daily
Timing	Every day at 1 AM
File Retention Policy	Files are retained for a maximum of 30 days in the sTMS SFTP server, after this period the files will be automatically deleted.
Possible Exception and Exception Handling	File loss or transfer failure due to sTMS internal exception. E.g., Timer service down, etc. Exception handling steps • sTMS support team shall be informed by email of the transfer failure. • sTMS support team checks the relevant logs to troubleshoot the cause of the failure. • If the issue is due to the transfer file could not be found in SFTP: ○ sTMS support team will manually rerun the timer job to trigger file generation and deposit the file in SP SFTP server.

2.3 Business Flow

2.3.1 Flow to Generate SG Secure TXT File

SN	Description
1	sTMS identifies the student’s enrollment status daily. Once the student is enrolled, sTMS shall locate the corresponding SP course info and student info.
2	sTMS generates a TXT file to consolidate the full-load data of students.
3	sTMS deposits the TXT file in SFTP server accordingly.

2.3.2 Business Rule

Please refer to [5. Interface Sample Files](#) for a better understanding of below business rules.

1. For certifiable/non-Certifiable courses process should send all active student, course status must be active, course intake status must be open for application, application closed, confirmed, commenced. For diploma courses, process should send to all active student before course end date.
2. Timer job should stop sending student data until expiry date + 7 days. For certifiable/non-Certifiable courses, sTMS pick expiry date from course end date, for diploma courses, sTMS pick expiry date from course end date unless candidate end date is populated.
3. For diploma courses students who have enrolment status as deferred, transferred, withdrawn, suspend, graduated and completed should stop sending to SG Secure.
4. If the student is active in more than one course, the system should send only 1 row of data based on the course code, semester, course intake No. until course end date + 7 days. The system should check if there are any course intakes currently running for this student on the current date. If yes, the system should send the valid date and expiry date of the specific course.
5. If the student is currently active in more than one course, the system should send the course with the highest/maximum end date. If there are currently no courses for this student on the current date, the system should send the earliest start date of the course in the future date.
6. Below is the mapping list of sTMS course category, course type and SP existing TMS course careers for sTMS to pick the course's valid date and expiry date.

Course Category	Course Type	SP Existing TMS Course Careers
Part-time Diploma (PTD)	Academic Full Qualification (FQ)	Diploma Course
	Academic Certificate (MC)	Diploma Course
	Academic Modular (SBMC)	Diploma Course
Specialist Diploma (SD)	Academic Full Qualification (FQ)	Diploma Course
	Academic Certificate (PDC)	Diploma Course
	Academic Modular (SBMC)	Diploma Course
Advanced Diploma (AD)	Academic Full Qualification (FQ)	Diploma Course
	Academic Certificate (PDC)	Diploma Course
	Academic Modular (SBMC)	Diploma Course
Diploma Conversion	Academic Full Qualification (FQ)	Diploma Course
	Academic Certificate (PDC)	Diploma Course
	Academic Modular (SBMC)	Diploma Course
Certifiable Courses	SkillsFuture Series	Certifiable/Non-Certifiable Course
	SkillsFuture Digital Confidence	Certifiable/Non-Certifiable Course
	SkillsFuture Digital Workplace	Certifiable/Non-Certifiable Course
	SkillsFuture Career Transition Programme (SCTP)	Certifiable/Non-Certifiable Course

	SGUnited Skills (SGUS) Programme	Certifiable/Non-Certifiable Course
	SGUnited Mid-Career Pathways (SGUP) Programme	Certifiable/Non-Certifiable Course
	Career Conversion Programme	Certifiable/Non-Certifiable Course
	Work-Learn Bootcamp	Certifiable/Non-Certifiable Course
	Certificate of Competency (COC) - SkillsFuture Series	Certifiable/Non-Certifiable Course
	General Education (GE)	Certifiable/Non-Certifiable Course
	Joint Industry Certificate (JIND)	Certifiable/Non-Certifiable Course
	Train The Trainer (TTT)	Certifiable/Non-Certifiable Course
	Workforce Skills Qualification (WSQ) Full Qualification	Certifiable/Non-Certifiable Course
	WSQ modular	Certifiable/Non-Certifiable Course
	Train-and-Place Programme (TnP)	Certifiable/Non-Certifiable Course
	Certifiable Short Courses	Short Course
Non-Certifiable Courses	SGUnited Skills (SGUS) Programme	Certifiable/Non-Certifiable Course
	Career Conversion Programme	Certifiable/Non-Certifiable Course
	Work-Learn Bootcamp	Certifiable/Non-Certifiable Course
	Work Study Certificate (WSC)	Certifiable/Non-Certifiable Course
	Seminars	Certifiable/Non-Certifiable Course
	SkillsFuture Series	Certifiable/Non-Certifiable Course
	Non-Certifiable Short Courses	Short Course
Work-Study Programmes	Work-Study Certificates Leading To Part-Time Diploma	Diploma Course
	Work-Study Post-Diploma Specialist Diploma	Diploma Course
	Work-Study Post-Diploma Leading To Specialist Diploma	Diploma Course
	Work-Study Post-Diploma Advanced Diploma	Diploma Course
	Workforce Skills Qualification (WSQ)	Diploma Course
	Work-Study Post-Diploma Diploma Conversion	Diploma Course

3. Interface Fields

Name	Description	Data Type	Remarks
Organization	Organization name	String	Default to SP
Department	Department/ School name	String	Default to PACE
Name	Student official name	String	
StudentID	Student Id	String	UAM Common Id
AccessNo	Access number for sTMS student barcode	String	Sample value: 1234567140922, 7 digits running number as access number & expired date in the form of ddmmyy The format to be confirmed, pending on the final solution of sTMS student barcode.
Email	Student personal email address	String	
Group	Group value: Permanent/ Contract/ Student	String	Default to STUDENT
ValidDate	Start date of course/ semester	String(\$DDMMYYYY)	Short courses and certifiable/non-Certifiable courses: Students pick from course start date. Diploma courses: course start date
ExpiryDate	End date of course/ semester	String(\$DDMMYYYY)	Short courses and certifiable/non-Certifiable courses: Students pick from course end date. Diploma courses: course end date unless candidate end date is populated.(Withdrawn, Suspended, Deferred, Transferred)

4. Interface Sample Files

List of SP SG Secure System scenario:



The below Interface sample file generated based on the SG Secure Scenario.xlsx described on specific time 03 Jan 2021 and 25 Jan 2021.

